

**Question:**

An 18-month-old boy presents to the emergency department for malaise. The boy’s parents report worsening fatigue for 3 days with associated irritability and anorexia. The patient’s newborn screening revealed a point mutation in the beta-globin gene but the patient has otherwise been healthy since birth. On physical exam, his temperature is 102.4°F (39.1°C), blood pressure is 78/42 mmHg, pulse is 124/min, and respirations are 32/min. The child is tired-appearing and difficult to soothe. Laboratory testing is performed and reveals the following: Serum: Na+: 137 mEq/L Cl-: 100 mEq/L K+: 4.4 mEq/L HCO3-: 24 mEq/L Urea nitrogen: 16 mg/dL Creatinine: 0.9 mg/dL Glucose: 96 mg/dL Leukocyte count: 19,300/mm^3 with normal differential Hemoglobin: 7.8 g/dL Hematocrit: 21% Mean corpuscular volume: 82 um^3 Platelet count: 324,000/mm^3 Reticulocyte index: 3.6% Which of the following is the most likely causative organism for this patient's presentation? Answer Choices: A. Haemophilus influenzae B. Neisseria meningitidis C. Salmonella D. Streptococcus pneumoniae

**Reasoning + Answer:**

<think>The patient had malaise , fatigue , and anorexia . The newborn screening revealed a point mutation in the beta-globin gene . The laboratory results showed abnormal hemoglobin and hematocrit levels . Haemophilus influenzae , Neisseria meningitidis , Salmonella , and Streptococcus pneumoniae were possible causative organisms. </think><answer>Haemophilus influenzae </answer>

